

ChintanJikkar_Individual-Visual-Analytics-Assignment-5.r

Chintan S. Jikkar

2025-10-14

##NAME : Chintan Samirbhai Jikkar [U01989702] | Visual Analytics

```
library(dplyr)
```

```
##
```

```
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
```

```
##
```

```
## filter, lag
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
## intersect, setdiff, setequal, union
```

```
library(ggplot2)
```

```
siva <- read.csv("/Users/chintanjikkar/Desktop/PACE/Fall 25/Visual Analytics/siva.csv")
```

```
str(siva)
```

```
## 'data.frame': 53815 obs. of 29 variables:
## $ xgra_n1clb_nbr : int 51407 23460 53417 14382 40539 53945 359
83 43669 29279 14254 ...
## $ Siva_Rental_Number : int 67041 56084 70279 15105 49797 71102 431
04 54673 33940 14950 ...
## $ rent_area_loc : int 156 204 181 1515 259 165 177 276 2167 9
53 ...
## $ Date_of_Survey : chr "5/18/2011" "2/5/2011" "6/14/2011" "1/4
/2010" ...
## $ Day_of_Week : chr "Wednesday" "Saturday" "Tuesday" "Monda
y" ...
## $ Time : chr "7:48:30" "22:06:37" "5:35:48" "23:58:5
6" ...
## $ Survey_Type : chr "SV Web Sol." "SV Web Sol." "SV Web Sol
." "SV Web Sol." ...
## $ Purpose_of_Rental : chr "Bus." "Leis. / Pers." "Bus." "Leis. /
Pers." ...
## $ Recom_mend_Siva : int 8 8 8 7 9 9 9 6 9 5 ...
## $ Staff_Courtesy : int 9 8 7 8 9 9 9 9 8 7 ...
## $ Speed_of_Service : int 8 8 8 7 9 9 9 8 9 5 ...
## $ Veh_Equip_Condition : int 9 5 8 8 9 9 9 9 8 8 ...
## $ Trans_Billing_as_Expected: int 9 8 8 8 9 9 9 9 9 7 ...
## $ Value_for_the_Money : int 9 7 8 8 9 9 9 6 7 7 ...
## $ Area : chr "01602 - LOVE FIELD AP TX" "07286 - VAL
LEJO CA OAP" "01850 - RICHMOND VA AP" "05743 - PICO CA OAP" ...
## $ loc_nm : chr "DALLAS LOVE FIELD" "VALLEJO HLE" "RICH
MOND INTL AP" "PICO HLE" ...
## $ ga_region_desc : chr "SOUTHWEST REGION" "WESTERN REGION" "MI
D ATLANTIC REGION" "WESTERN REGION" ...
## $ xgra_ckot_ts : chr "5/15/2011" "1/31/2011" "6/12/2011" "12
/23/2009" ...
## $ xgra_ckin_ts : chr "5/17/2011" "2/3/2011" "6/13/2011" "1/3
/2010" ...
## $ xgra_vclass_reserv : chr "C" "A" "F" "D" ...
## $ xgra_veh_class : chr "Q4" "B" "YF" "YF" ...
## $ rent_loc_type : chr "AP" "OFF AP" "AP" "OFF AP" ...
## $ cust_tier_code : chr "FG" "N1" "RG" "RG" ...
## $ booking_channel_code : chr "SIVA.COM" "SIVA.COM" "SIVA.COM" "SIVA.
COM" ...
## $ col34_total_charges : num 247.3 128 75.8 468.5 42.8 ...
## $ col38_currency : chr "USD" "USD" "USD" "USD" ...
## $ Total_charge_USD : num 247.3 128 75.8 468.5 42.8 ...
## $ Survey_checkout_diff : int 2 3 2 2 2 4 2 2 6 2 ...
## $ booking_channel_dummy : int 1 1 1 1 1 0 0 0 1 1 ...
```

```
head(siva, 10)
```

```
##      xgra_n1clb_nbr Siva_Rental_Number rent_area_loc Date_of_Survey Day_of_Week
## 1          51407          67041          156      5/18/2011      Wednesday
## 2          23460          56084          204      2/5/2011      Saturday
## 3          53417          70279          181      6/14/2011      Tuesday
## 4          14382          15105         1515      1/4/2010      Monday
## 5          40539          49797          259     12/1/2010     Wednesday
## 6          53945          71102          165      6/29/2011     Wednesday
## 7          35983          43104          177      9/23/2010      Thursday
## 8          43669          54673          276      1/23/2011      Sunday
## 9          29279          33940         2167      7/6/2010      Tuesday
## 10         14254          14950          953      1/7/2010      Thursday

##      Time Survey_Type Purpose_of_Rental Recom_mend_Siva Staff_Courtesy
## 1    7:48:30 SV Web Sol.          Bus.           8           9
## 2   22:06:37 SV Web Sol.    Leis. / Pers.        8           8
## 3    5:35:48 SV Web Sol.          Bus.           8           7
## 4   23:58:56 SV Web Sol.    Leis. / Pers.        7           8
## 5    8:24:39 SV Web Sol.    Leis. / Pers.        9           9
## 6    5:34:06 SV Web Sol.          Bus.           9           9
## 7    6:24:40 SV Web Sol.          Bus.           9           9
## 8    9:48:08 SV Web Sol.          Bus.           6           9
## 9   14:38:54 SV Web Sol.          Bus.           9           8
## 10   5:49:29 SV Web Sol.          Bus.           5           7

##      Speed_of_Service Veh_Equip_Condition Trans_Billing_as_Expected
## 1           8           9           9
## 2           8           5           8
## 3           8           8           8
## 4           7           8           8
## 5           9           9           9
## 6           9           9           9
## 7           9           9           9
## 8           8           9           9
## 9           9           8           9
## 10          5           8           7

##      Value_for_the_Money Area
## 1           9      01602 - LOVE FIELD AP TX
## 2           7      07286 - VALLEJO CA OAP
## 3           8      01850 - RICHMOND VA AP
```

## 4	8	05743 - PICO CA OAP
## 5	9	07787 - NEWPORT RI OAP
## 6	9	01450 - ATLANTA AP GA
## 7	9	05426 - ELIZABETHTOWN: KY OAP
## 8	6	02170 - SALT LAKE CITY UT AP
## 9	7	07275 - WALSH RD. SANTA CLARA CA OAP
## 10	7	07781 - BEDFORD MA OAP
##	loc_nm	ga_region_desc xgra_ckot_ts xgra_ckin_ts
## 1	DALLAS LOVE FIELD	SOUTHWEST REGION 5/15/2011 5/17/2011
## 2	VALLEJO HLE	WESTERN REGION 1/31/2011 2/3/2011
## 3	RICHMOND INTL AP	MID ATLANTIC REGION 6/12/2011 6/13/2011
## 4	PICO HLE	WESTERN REGION 12/23/2009 1/3/2010
## 5	NEWPORT HLE	NORTHEAST REGION 11/29/2010 11/30/2010
## 6	ATLANTA-HARTSFIELD INTL	SOUTHEAST REGION 6/20/2011 6/26/2011
## 7	ELIZABETHTOWN HLE	CENTRAL REGION 9/19/2010 9/22/2010
## 8	SALT LAKE CITY INTL AP	WEST CENTRAL REGION 1/17/2011 1/22/2011
## 9	INTEL (SC12)	WESTERN REGION 6/30/2010 7/1/2010
## 10	JET AVIATION	NORTHEAST REGION 1/3/2010 1/6/2010
##	xgra_vclass_reserv	xgra_veh_class rent_loc_type cust_tier_code
## 1	C	Q4 AP FG
## 2	A	B OFF AP N1
## 3	F	YF AP RG
## 4	D	YF OFF AP RG
## 5	F	A OFF AP RG
## 6	R	YR AP RG
## 7	Q4	Q4 OFF AP N1
## 8	F	T AP N1
## 9	B	C OFF AP RG
## 10	D	F OFF AP RG
##	booking_channel_code	col34_total_charges col38_currency Total_charge_US
D		
## 1	SIVA.COM	247.29 USD 247.2
9		
## 2	SIVA.COM	128.04 USD 128.0
4		
## 3	SIVA.COM	75.85 USD 75.8
5		
## 4	SIVA.COM	468.51 USD 468.5
1		
## 5	SIVA.COM	42.84 USD 42.8
4		
## 6	800#	107.92 USD 107.9
2		
## 7	LOCAL RES	224.68 USD 224.6
8		
## 8	GDSB	387.63 USD 387.6
3		
## 9	SIVA.COM	45.08 USD 45.0
8		
## 10	SIVA.COM	229.10 USD 229.1

```
0
##      Survey_checkout_diff booking_channel_dummy
## 1              2              1
## 2              3              1
## 3              2              1
## 4              2              1
## 5              2              1
## 6              4              0
## 7              2              0
## 8              2              0
## 9              6              1
## 10             2              1
```

QUESTION 1

##Subset 1 selects all rental records for Chicago O'Hare Airport.

```
sub1 <- filter(siva, loc_nm == "CHICAGO OHARE AP")
head(sub1, 10)
```

##	xgra_n1clb_nbr	Siva_Rental_Number	rent_area_loc	Date_of_Survey	Day_of_Week
## 1	55478	73479	44	7/17/2011	Sun
## 2	26534	30290	44	6/6/2010	Sun
## 3	39462	75279	44	8/13/2011	Satur
## 4	36018	43152	44	10/4/2010	Mon
## 5	33133	39033	44	8/19/2010	Thurs
## 6	2434	2447	44	8/25/2009	Tues
## 7	16125	17113	44	1/25/2010	Mon
## 8	1082	1084	44	8/14/2009	Fri
## 9	23472	26240	44	5/3/2010	Mon
## 10	11162	31532	44	6/6/2010	Sun

##	Time	Survey_Type	Purpose_of_Rental	Recom_mend_Siva	Staff_Courtesy
## 1	12:37:48	SV Web Sol.	Bus.	6	8
## 2	9:58:44	SV Web	Leis. / Pers.	1	6
## 3	12:21:13	SV Web Sol.	Bus.	9	9
## 4	20:02:30	SV Web Sol.	Bus.	4	5
## 5	13:14:40	SV Web	Bus.	9	9
## 6	12:09:08	SV Web	Bus.	9	9
## 7	15:42:31	SV Phone	Leis. / Pers.	2	NA
## 8	8:21:43	SV Web Sol.	Bus.	9	9
## 9	12:52:05	SV Web Sol.	Leis. / Pers.	8	9
## 10	16:33:00	SV Web Sol.	Leis. / Pers.	2	8

##	Speed_of_Service	Veh_Equip_Condition	Trans_Billing_as_Expected
## 1	9	7	9
## 2	8	9	9
## 3	9	9	9
## 4	5	6	6
## 5	8	8	9
## 6	9	8	9
## 7	NA	NA	NA
## 8	9	8	9
## 9	9	9	9
## 10	8	7	0

##	Value_for_the_Money	Area	loc_nm
----	---------------------	------	--------

## 1	6 02201 - CHICAGO O'HARE AP IL CHICAGO OHARE AP
## 2	7 02201 - CHICAGO O'HARE AP IL CHICAGO OHARE AP
## 3	9 02201 - CHICAGO O'HARE AP IL CHICAGO OHARE AP
## 4	4 02201 - CHICAGO O'HARE AP IL CHICAGO OHARE AP
## 5	7 02201 - CHICAGO O'HARE AP IL CHICAGO OHARE AP
## 6	6 02201 - CHICAGO O'HARE AP IL CHICAGO OHARE AP
## 7	NA 02201 - CHICAGO O'HARE AP IL CHICAGO OHARE AP
## 8	8 02201 - CHICAGO O'HARE AP IL CHICAGO OHARE AP
## 9	9 02201 - CHICAGO O'HARE AP IL CHICAGO OHARE AP
## 10	8 02201 - CHICAGO O'HARE AP IL CHICAGO OHARE AP
##	ga_region_desc xgra_ckot_ts xgra_ckin_ts xgra_vclass_reserv xgra_veh_cl
ass	
## 1	CENTRAL REGION 7/10/2011 7/16/2011 C
C	
## 2	CENTRAL REGION 5/25/2010 5/29/2010 C
D	
## 3	CENTRAL REGION 8/2/2011 8/12/2011 B
YB	
## 4	CENTRAL REGION 9/19/2010 9/23/2010 C
C	
## 5	CENTRAL REGION 8/15/2010 8/18/2010 C
D4	
## 6	CENTRAL REGION 8/23/2009 8/24/2009 F
F	
## 7	CENTRAL REGION 1/21/2010 1/23/2010 F
G	
## 8	CENTRAL REGION 8/9/2009 8/13/2009 F
YF	
## 9	CENTRAL REGION 4/16/2010 4/26/2010 T
S4	
## 10	CENTRAL REGION 6/3/2010 6/5/2010 E6
YF	
##	rent_loc_type cust_tier_code booking_channel_code col34_total_charges
## 1	AP FG GDSC 272.31
## 2	AP N1 LINK FR INS CO 177.46
## 3	AP FG GDSB 425.04
## 4	AP N1 GDSB 298.79
## 5	AP PC SIVA.COM 220.31
## 6	AP FG GDSB 146.37
## 7	AP RG SIVA.COM 84.06
## 8	AP FG 0 259.40
## 9	AP N1 SIVA.COM 1803.15
## 10	AP FG SIVA.COM 51.39
##	col38_currency Total_charge_USD Survey_checkout_diff booking_channel_du
mmy	
## 1	USD 272.31 2
0	
## 2	USD 177.46 9
0	
## 3	USD 425.04 2

0				
## 4	USD	298.79		12
0				
## 5	USD	220.31		2
1				
## 6	USD	146.37		2
0				
## 7	USD	84.06		3
1				
## 8	USD	259.40		2
0				
## 9	USD	1803.15		8
1				
## 10	USD	51.39		2
1				

##Subset 2 extracst only the 3 columns mentioned from the observations.

```
sub2 <- select(siva, Area, ga_region_desc, Total_charge_USD)
```

```
head(sub2, 10)
```

##		Area	ga_region_desc	Total_charge_U
SD				
## 1		01602 - LOVE FIELD AP TX	SOUTHWEST REGION	247.
29				
## 2		07286 - VALLEJO CA OAP	WESTERN REGION	128.
04				
## 3		01850 - RICHMOND VA AP	MID ATLANTIC REGION	75.
85				
## 4		05743 - PICO CA OAP	WESTERN REGION	468.
51				
## 5		07787 - NEWPORT RI OAP	NORTHEAST REGION	42.
84				
## 6		01450 - ATLANTA AP GA	SOUTHEAST REGION	107.
92				
## 7		05426 - ELIZABETHTOWN: KY OAP	CENTRAL REGION	224.
68				
## 8		02170 - SALT LAKE CITY UT AP	WEST CENTRAL REGION	387.
63				
## 9	07275 - WALSH RD. SANTA CLARA CA OAP		WESTERN REGION	45.
08				
## 10		07781 - BEDFORD MA OAP	NORTHEAST REGION	229.
10				

```
##Subset 3 adds a High_Recommendation flag showing if a customer rated >=8.
siva$High_Recommendation <- ifelse(siva$Recom_mend_Siva >= 8, "Yes", "No")
sub3 <- select(siva, Siva_Rental_Number, Recom_mend_Siva, High_Recommendation
)
head(sub3, 10)
```

##	Siva_Rental_Number	Recom_mend_Siva	High_Recommendation
## 1	67041	8	Yes
## 2	56084	8	Yes
## 3	70279	8	Yes
## 4	15105	7	No
## 5	49797	9	Yes
## 6	71102	9	Yes
## 7	43104	9	Yes
## 8	54673	6	No
## 9	33940	9	Yes
## 10	14950	5	No

##Subset 4 counts the number of rentals at each location in the dataset.

```
sub4 <- siva %>%  
  group_by(loc_nm) %>%  
  summarise(Count = n())  
head(sub4, 10)
```

```
## # A tibble: 10 × 2
```

	loc_nm	Count
	<chr>	<int>
## 1	10TH STREET HLE	58
## 2	12 EAST 13TH ST (E)	35
## 3	126 W. 55TH ST. (W)	40
## 4	17TH & BEN FRANKLIN HLE	26
## 5	210 W. 77TH ST. (W)	36
## 6	214 W. 95TH ST. (W)	38
## 7	222 E. 40TH ST. (E)	42
## 8	270 W 60TH ST	12
## 9	310 E. 48TH ST. (E)	32
## 10	323 W. 34TH ST. (W)	24

##Subset 5 calculates the average staff courtesy score for each customer tier group.

```
sub5 <- siva %>%  
  group_by(cust_tier_code) %>%  
  summarise(Avg_Staff_Courtesy = mean(Staff_Courtesy, na.rm = TRUE))  
head(sub5, 10)
```

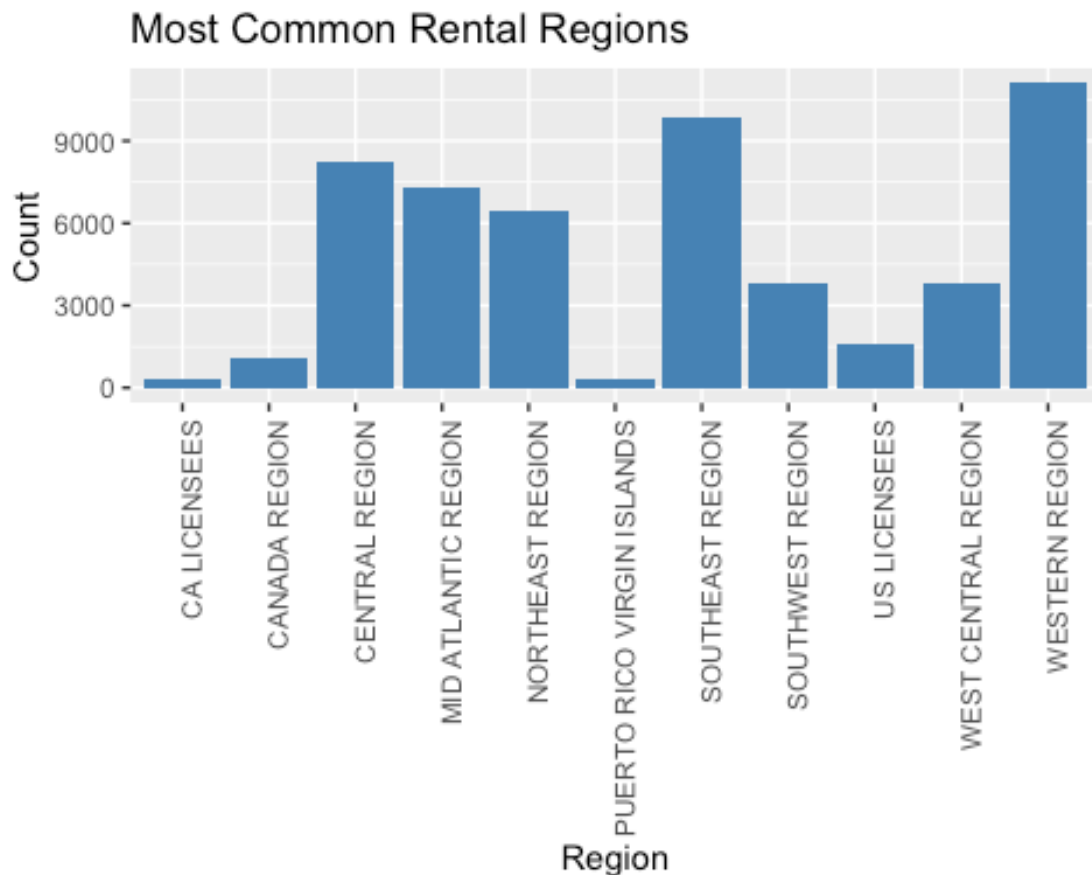
```
## # A tibble: 8 × 2
```

```
##   cust_tier_code Avg_Staff_Courtesy  
##   <chr>          <dbl>  
## 1 FG            8.13  
## 2 N1            8.11  
## 3 PC            8.25  
## 4 PL            8.50  
## 5 PS            8.67  
## 6 RG            8.09  
## 7 VP            7.25  
## 8 <NA>          7.66
```

QUESTION 2

##Barplot of Most Common Rental Locations

```
ggplot(siva, aes(x = ga_region_desc)) +  
  geom_bar(fill = "steelblue") +  
  labs(title = "Most Common Rental Regions", x = "Region", y = "Count") +  
  theme(axis.text.x = element_text(angle = 90, hjust = 1))
```

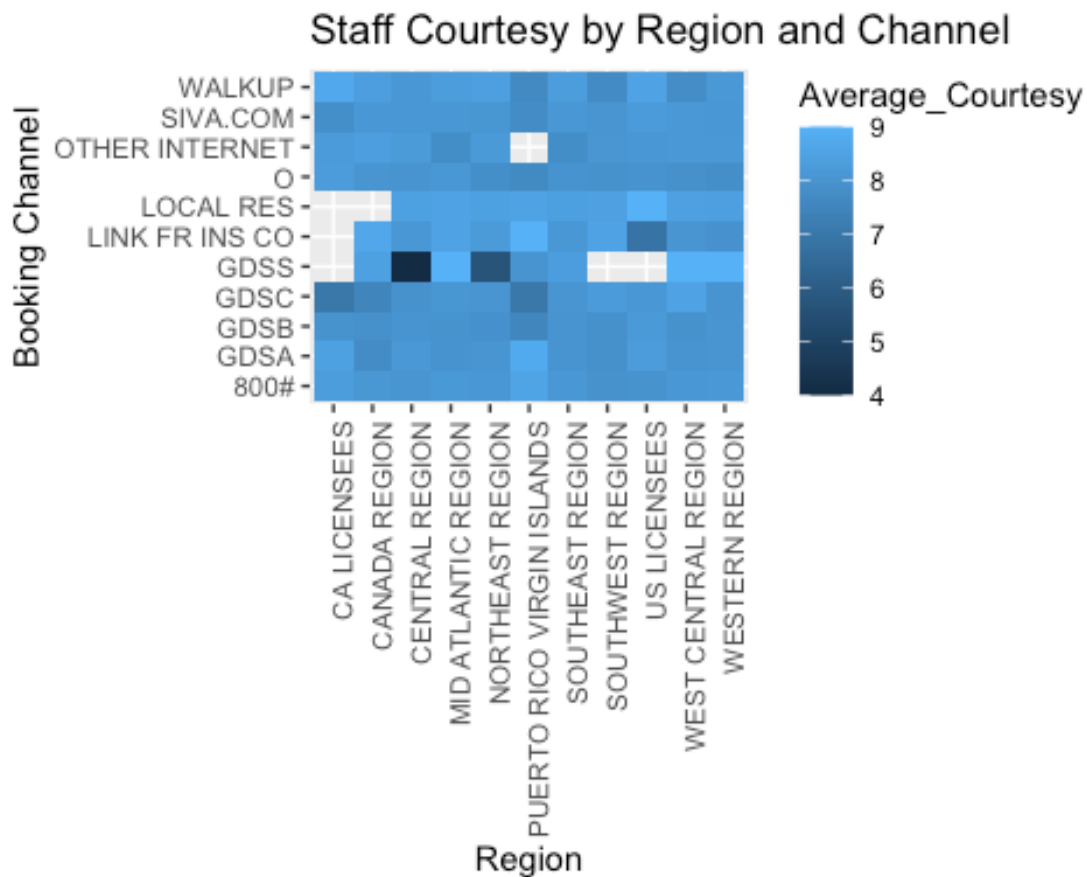


#The Western Region has the highest rental count, highlighting its role as Siva's busiest market.

#Regions like Puerto Rico Virgin Islands and CA Licensee's have very few rentals, suggesting either low demand or limited operations there.

##Heatmap of Satisfaction by Region and Booking Channel

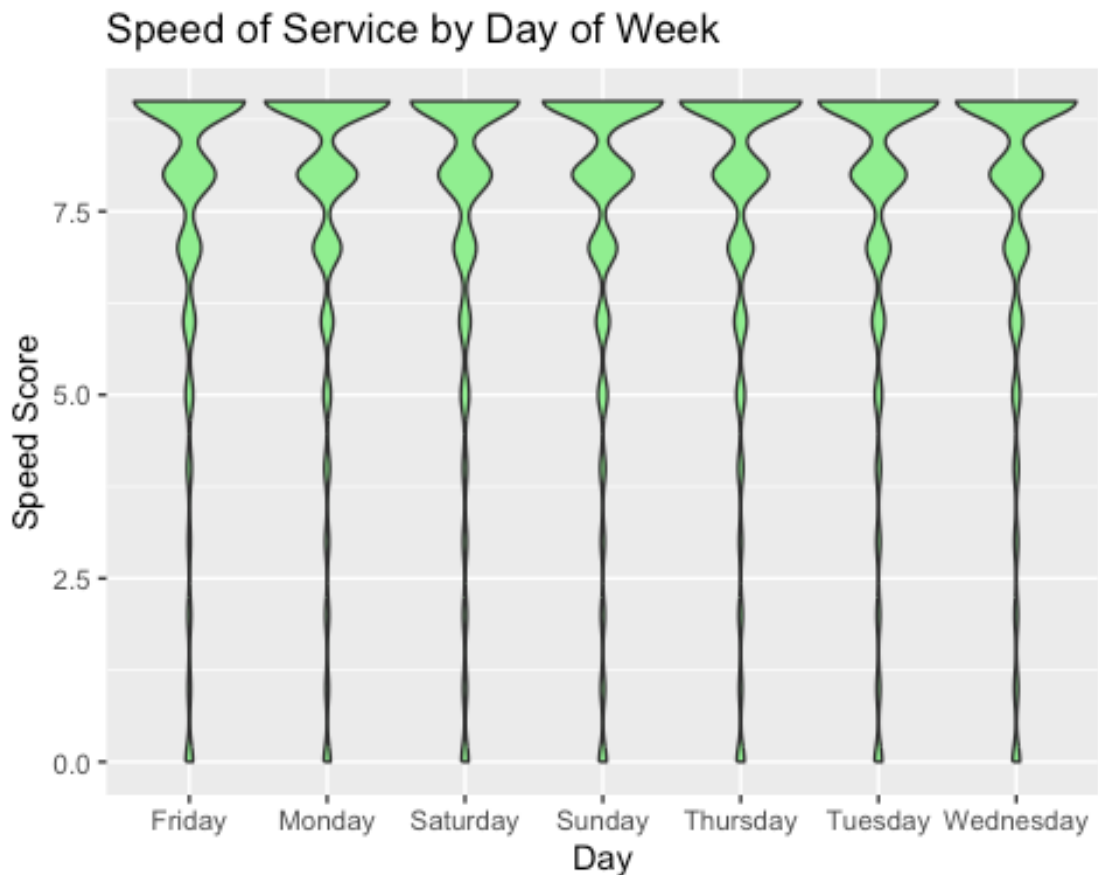
```
heat <- siva %>%
  dplyr::group_by(ga_region_desc, booking_channel_code) %>%
  dplyr::summarise(Average_Courtesy = mean(Staff_Courtesy, na.rm = TRUE), .groups = "drop")
ggplot(heat, aes(x = ga_region_desc, y = booking_channel_code, fill = Average_Courtesy)) +
  geom_tile() +
  labs(title = "Staff Courtesy by Region and Channel", x = "Region", y = "Booking Channel") +
  theme(axis.text.x = element_text(angle = 90, hjust = 1))
```



#Courtesy ratings are generally high across most regions and booking channels, but there are noticeable pockets where courtesy dips, especially for less common channels in specific regions.

#SIVA.COM and Walk-up channels deliver consistently high courtesy scores in most regions, suggesting direct bookings lead to a better customer service experience.

```
##Violin Plot for Speed of Service by Day of Week
ggplot(filter(siva, !is.na(Speed_of_Service)),
  aes(x = Day_of_Week, y = Speed_of_Service)) +
  geom_violin(fill = "lightgreen") +
  labs(title = "Speed of Service by Day of Week", x = "Day", y = "Speed Score
")
```

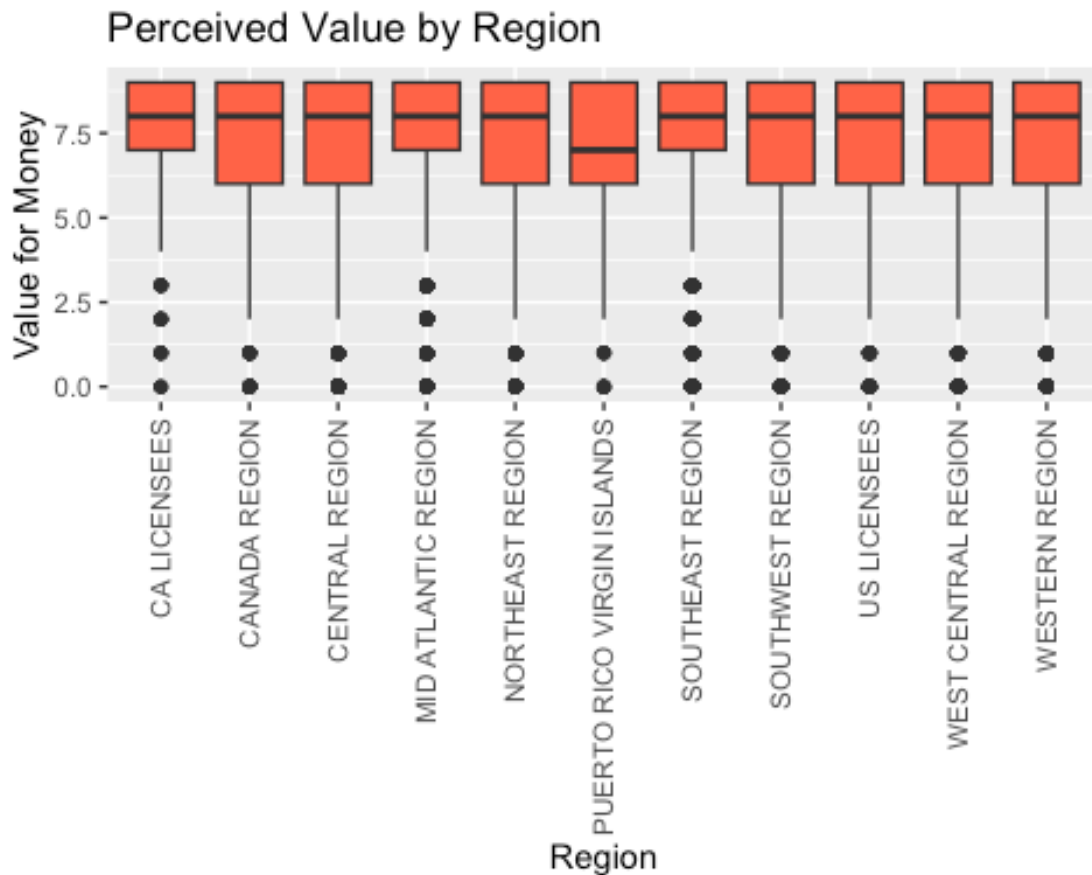


#Speed of service is consistently high across all days of the week, with no significant variation between weekdays and weekends.

#There is no evidence of a "slow day," meaning service standards are well maintained throughout the week.

##Boxplot of Value for Money by Rental Region

```
ggplot(filter(siva, !is.na(Value_for_the_Money)),  
  aes(x = ga_region_desc, y = Value_for_the_Money)) +  
  geom_boxplot(fill = "tomato") +  
  labs(title = "Perceived Value by Region", x = "Region", y = "Value for Mone  
y") +  
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust = 1))
```

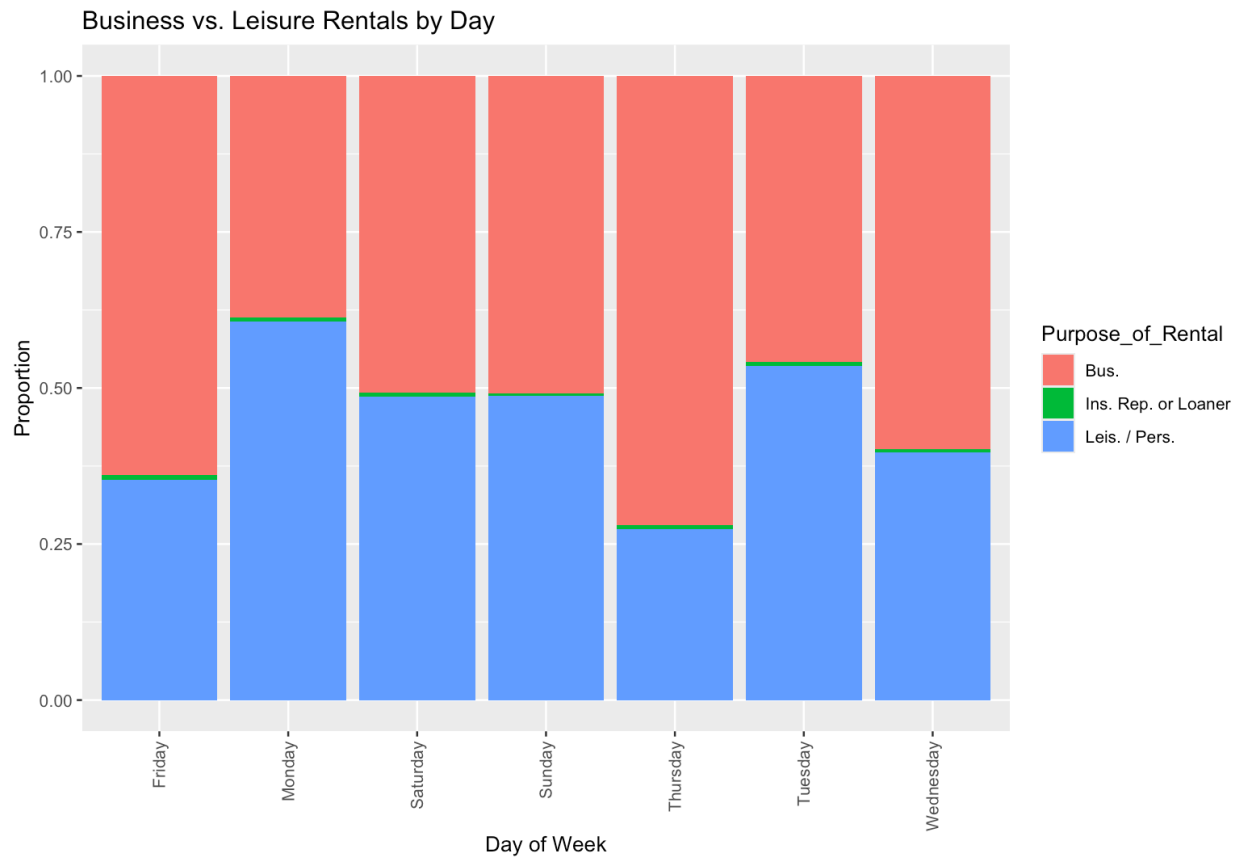


#Perceived value scores are consistently high across all regions, with most medians near the top of the scale.

#There is no significant gap in value for money by region, meaning Siva customers generally feel they are getting good value regardless of where they rent.

##Stacked Bar Plot for Frequency of Business vs. Leisure Rentals by Day of Week

```
ggplot(siva, aes(x = Day_of_Week, fill = Purpose_of_Rental)) +
  geom_bar(position = "fill") +
  labs(title = "Business vs. Leisure Rentals by Day", x = "Day of Week", y =
"Proportion") +
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust = 1))
```



#Leisure rentals dominate weekends, while business rentals are more frequent during weekdays, especially on Thursdays and Fridays.

#This pattern suggests that business travelers rent more on workdays, while personal and leisure rentals spike around weekends.

THANK YOU